

Simplified Quant Trading System

Quantitative Modelling Program, Final Project Report

1. Project Overview

The aim of this project is to build a small but complete quant trading workflow in Python and use it to test two simple trading rules on real market data. We are not trying to build something profitable or production grade. We are trying to understand what each piece of a trading system actually does and what kind of bookkeeping you have to get right so the results you report are not a lie.

The whole thing is broken into five layers that match the way the course presented the material:

1. A market data layer that fetches and cleans OHLCV data.
2. An analytics layer that turns prices into returns, moving averages, rolling volatility, and Bollinger bands.
3. A signal engine that takes those indicators and produces 0/1 BUY signals for two strategies.
4. An execution layer that simulates market and limit orders against a simple bid/ask spread.
5. An evaluation layer that runs a vectorised backtest, applies transaction costs, and reports the standard performance metrics.

All of this logic lives inside `src/` as separate Python modules. The notebook `notebook/project_notebook.ipynb` imports those modules and walks through the four problem statements end to end, with all chart outputs saved into `plots/`. The cleaned data is in `data/`. This report summarises what was built and what we found.

The most important habit we tried to enforce is that every backtest uses the position from the *previous* bar, not the current one. This is the `signal.shift(1)` step that the lectures kept calling out. Without it the strategy can effectively peek at the close it is using to make a decision, which makes the backtest look much better than it should.

2. Dataset Description

We used SPY, the SPDR S&P 500 ETF, downloaded directly from Yahoo Finance using the `yfinance` library. SPY tracks the S&P 500 index and is one of the most liquid instruments on the US market, so the daily OHLCV data is clean and there are no real gaps to worry about. It is also the benchmark almost every active strategy is judged against, so running our strategies on it makes the buy-and-hold comparison required by Problem Statement 4 directly meaningful.

The window we picked is 1 January 2018 to 31 December 2024. That is about seven years, or roughly 1,760 trading days. We picked this window on purpose because it contains every kind of regime an equity index can go through:

- A choppy late 2018 with the Q4 sell off.
- A trending 2019 bull market.
- A short violent crash in March 2020 caused by COVID.
- A very fast recovery and rally through 2020 and 2021.
- A grinding 2022 bear market driven by rate hikes.
- A strong bull rally through 2023 and 2024.

A shorter window would have given us only one type of regime, and we would not have been able to talk about how each strategy behaves when the market changes. The columns are the standard six the project asks for: Date as the index, then Open, High, Low, Close, and Volume.

For cleaning we did three things. First, drop any row that is fully empty. Second, forward fill the remaining gaps using the previous day's value (there are almost none on SPY, but the code handles it). Third, drop any row where one of the price columns is zero or negative because that would be a corrupted record. After cleaning there were no missing values left in the dataset and every row passed the sanity check that High is greater than or equal to Low and contains both Open and Close.

3. Indicators Used

We picked five indicators plus Bollinger bands. They are the same set the course covered in Lectures 3 and 4. We chose them because each captures a different feature of the price series, and our two signal rules need a subset of them as inputs.

Indicator	Formula	What it tells us
Simple return	<code>Close.pct_change()</code>	One day price movement in percent.
Cumulative return	<code>(1 + Return).cumprod() - 1</code>	Total return from day 0 to day t.
Short MA	<code>Close.rolling(20).mean()</code>	Smoothed price over the last month.
Long MA	<code>Close.rolling(50).mean()</code>	Smoothed price over the last two and a half months.
Rolling volatility	<code>Return.rolling(20).std()</code>	How noisy the returns have been recently.
Volume average	<code>Volume.rolling(20).mean()</code>	Normal trading activity baseline.

Indicator	Formula	What it tells us
Bollinger bands	$MA20 \pm 2 * std20$	Range that price spends most of its time inside.

The window choices (20 and 50 days) are the conventional ones used in the lecture material and in most retail technical analysis. We did not optimise them on the SPY data because we wanted to avoid in sample overfitting. Twenty trading days is roughly one calendar month and fifty is roughly two and a half months, which lines up with how practitioners think about short and intermediate trends.

The price together with both moving averages looks like this:

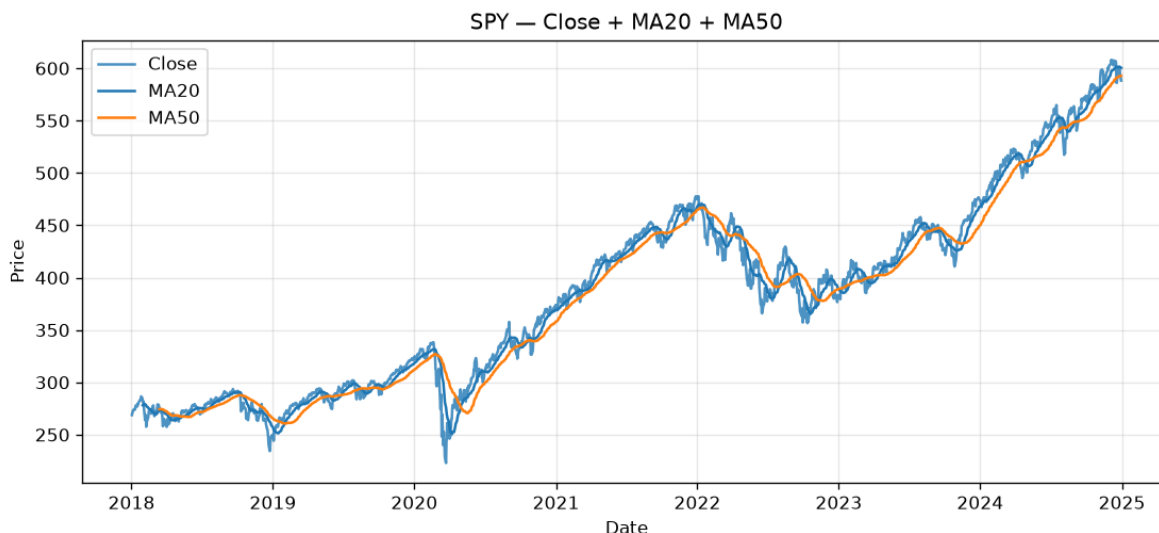


Figure 1: SPY close price with MA20 and MA50 overlays.

You can already read off the regimes by eye. The COVID crash in March 2020 is the cliff in the middle. The 2022 bear market is the gentle slope downwards in the middle right. The 2023 to 2024 rally is the steep climb at the end.

And the rolling 20 day volatility, which we use later when we talk about why the strategies behave the way they do:

The spike in March 2020 is the COVID crash. Volatility went from roughly one percent per day to almost six percent per day in two weeks. We will come back to this when we discuss why MA crossover struggled around that time.

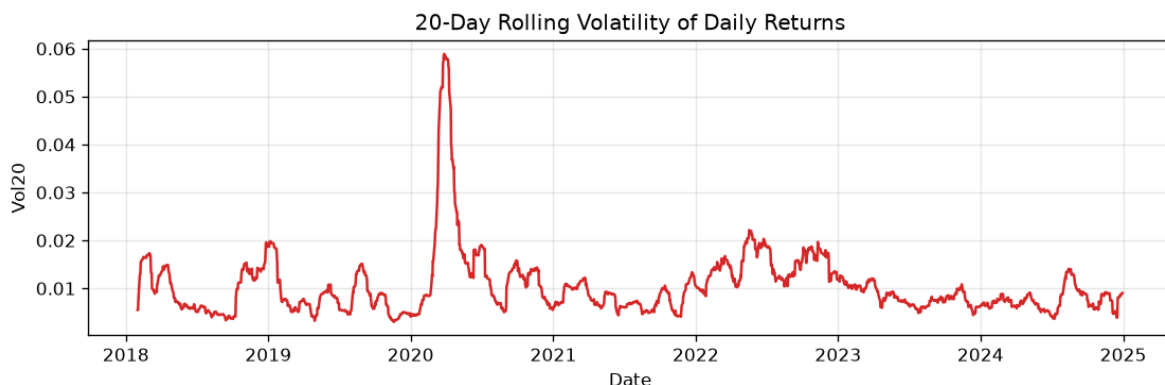


Figure 2: 20-day rolling standard deviation of daily SPY returns.

4. Trading Strategy Logic

We implemented two strategies plus a buy and hold benchmark. Both active strategies are long only and produce a 0/1 signal, where 1 means we want to be long for the next bar and 0 means we want to be flat. The crucial step is that we compute a Position column as `Signal.shift(1)`, which means today's position is whatever the signal said at yesterday's close. This is the no look ahead rule.

4.1 Strategy A: Moving Average Crossover

The rule is the standard textbook one. We go long when the 20 day moving average is above the 50 day moving average. We go flat when the 20 day moving average crosses back below. The intuition is that when the short term average is higher than the long term one the market is trending up, and we want to be in that trend.

The signal markers on the price chart show every entry and exit:

Notice how the strategy waits a while before entering after the COVID crash. The 20 day average has to climb back above the 50 day average, and because both averages are themselves smoothed, this takes a while. That delay costs us a lot of the recovery upside. This is the classic weakness of trend following on V shaped reversals.

4.2 Strategy B: Mean Reversion

This one is the opposite philosophy. We assume the price has a mean it tends to come back to, and we try to buy dips. Specifically, we go long when the close falls more than 3 percent below the 20 day moving average. We flatten the position as soon as the price rises back above the moving average.

A 3 percent threshold sounds small but on a large cap ETF like SPY it is actually material. Most days SPY moves less than 1 percent. A 3 percent drop in one or two sessions is a small panic. So the strategy is essentially betting on small panics resolving themselves, which is historically true for SPY.

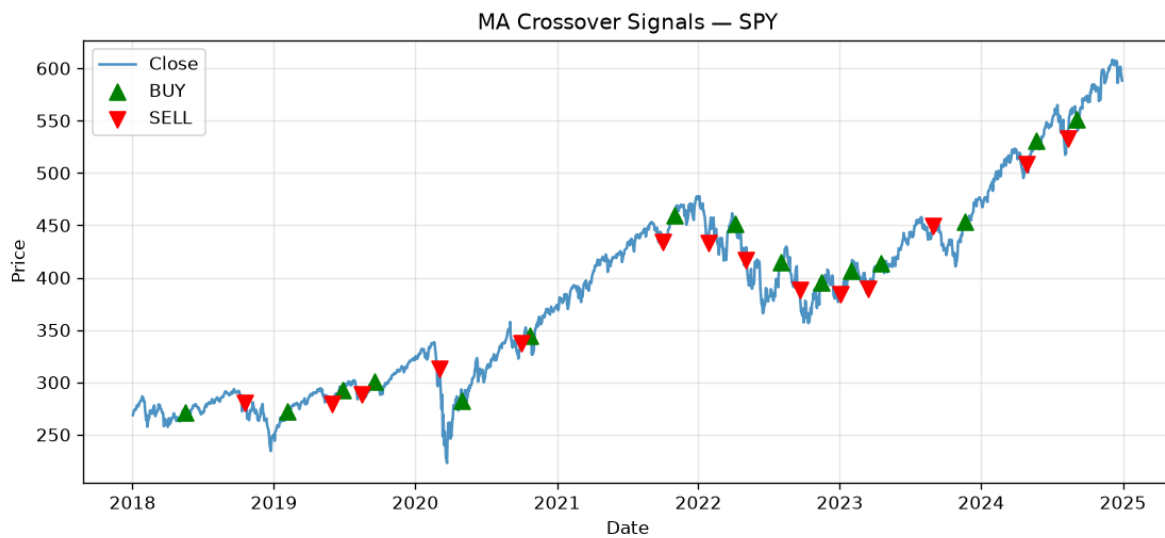


Figure 3: MA crossover signal markers on SPY. Green triangles are BUY (signal goes from 0 to 1), red triangles are SELL (1 to 0).

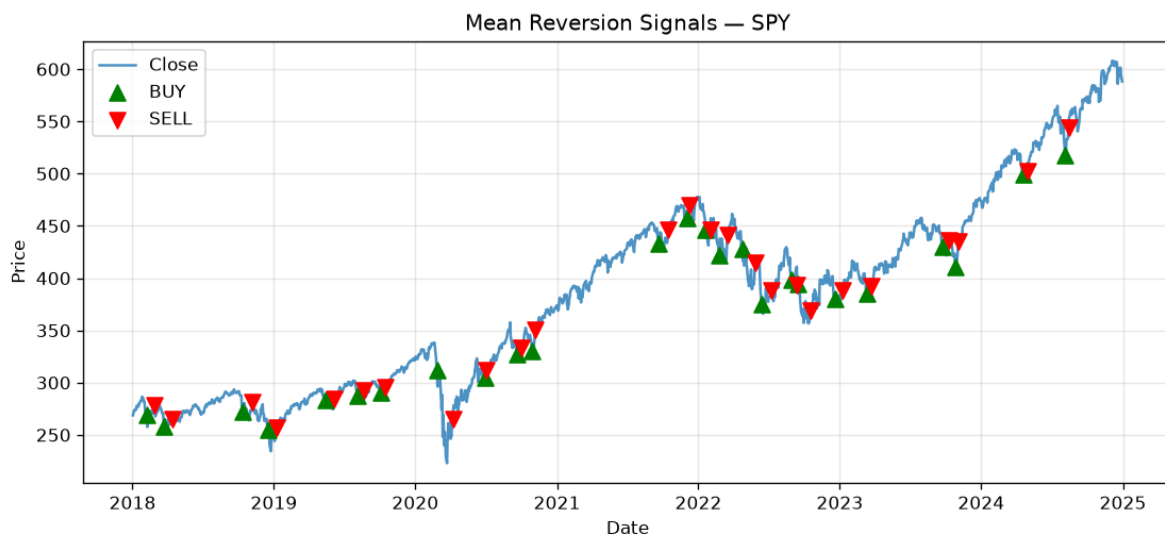


Figure 4: Mean reversion signal markers. The strategy fires whenever the close drops more than 3 percent below MA20.

4.3 Benchmark: Buy and Hold

A constant long signal from the first bar. This is required by PS4 as the passive baseline that any active strategy has to beat. In a strong bull regime it is a very hard benchmark.

5. Execution Logic

Problem Statement 3 asks us to simulate market and limit orders with a simple spread model. We put a fixed half spread of 5 cents around the close price:

$\text{Bid} = \text{Close} - \text{spread} / 2$

$\text{Ask} = \text{Close} + \text{spread} / 2$

For SPY this is about 1 basis point at the levels SPY trades at, which is roughly what the real spread looks like during regular trading hours. The four order types behave like this:

Order type	Fills at	Condition
Market BUY	Ask	Always (assuming the bar exists).
Market SELL	Bid	Always.
Limit BUY	Ask	Only if Ask is less than or equal to the limit price.
Limit SELL	Bid	Only if Bid is greater than or equal to the limit price.

These rules are coded in `src/execution.py`. The `simulate_orders` function walks through the bars in order. For a market order it always fills on the bar after it was submitted. For a limit order it keeps checking each bar until either the price condition is met or the backtest ends without a fill. This is what gives us the execution uncertainty the project asks us to demonstrate. In the notebook section 3.2 we submit a limit buy 1 percent below the average close and a limit sell 2 percent above. The buy fills early but the sell never does because the price keeps trending and never reaches the level.

On top of the spread we deduct an explicit transaction cost of 10 basis points and slippage of 5 basis points on every signal flip. This matches the staged backtest from Lecture 5 Part 2. A round trip therefore costs us at least 30 basis points (15 on the entry and 15 on the exit), plus the spread that the fill price itself includes. For a strategy that flips often this adds up.

6. Results and Analysis

This is the main results table. The numbers are for the full SPY window with 10 bps cost and 5 bps slippage applied on every signal flip.

Metric	MA Crossover	Mean Reversion	Buy and Hold
Total trades	15	25	1
Winning trades	9	19	1
Losing trades	6	6	0
Win rate	60.0%	76.0%	100%
Avg return per trade	2.64%	0.71%	91.6%
Total return	38.1%	7.1%	118.5%
Annualised return	5.4%	2.0%	13.1%
Annualised volatility	12.1%	14.2%	19.5%
Sharpe ratio ($R_f = 0$)	0.44	0.14	0.67
Max drawdown	30.9%	28.9%	34.1%

Three things jump out. First, buy and hold wins on every metric that matters. It has the highest absolute return, the highest Sharpe ratio, and even though its drawdown is the largest, the gap is small compared to how much more upside it captured. Second, MA Crossover is a kind of defensive watered down version of buy and hold. It captured about a third of the total return at about two thirds of the volatility, with a slightly lower drawdown. Third, mean reversion only made 7 percent over seven years even though it has a 76 percent win rate. This is the classic “high hit rate, tiny edge per trade, missed the big move” pattern.

The equity curves make this very visual:

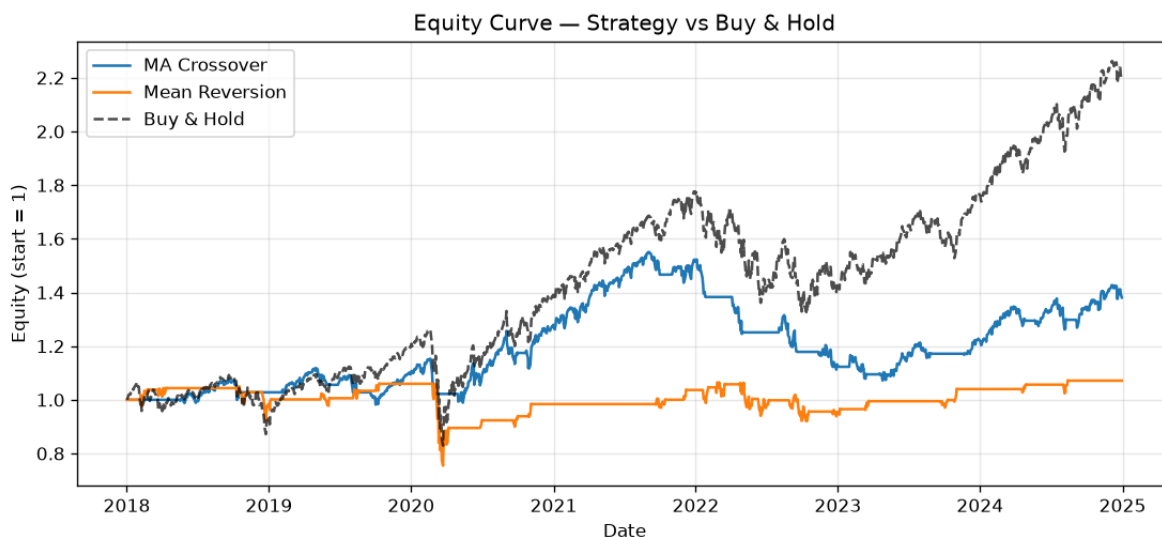


Figure 5: Equity curves: MA Crossover and Mean Reversion against the buy and hold benchmark on SPY. Starts at 1.0.

The dashed black line is buy and hold. You can see how cleanly it pulls away from both active strategies in late 2020 and again in 2023. The MA Crossover line (blue) flattens out during the

choppy 2022 period because the strategy keeps getting in and out. Mean reversion (orange) barely moves at all after 2021 because every time price recovers and crosses MA20 the strategy exits with a small profit and then waits for the next dip, which never comes during a bull market.

The drawdown plot of the better active strategy shows how the worst losing period actually felt:

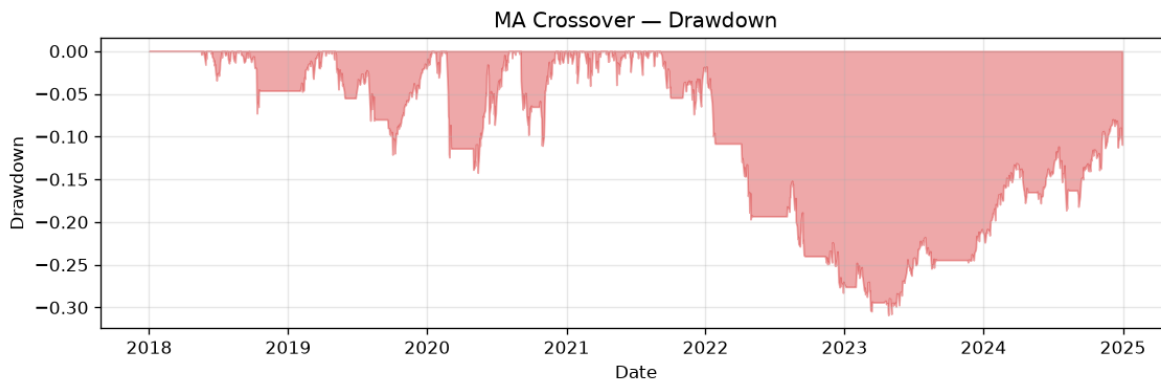


Figure 6: Drawdown chart for the better Sharpe active strategy (MA Crossover). Plotted as percentage below the running peak equity.

The deepest drawdown is in early 2020 when COVID hit and again in 2022. Both correspond to the regimes where MA Crossover is most exposed: fast reversals it cannot front run, and a year where the short and long averages keep flipping.

The daily return distribution of the strategy:

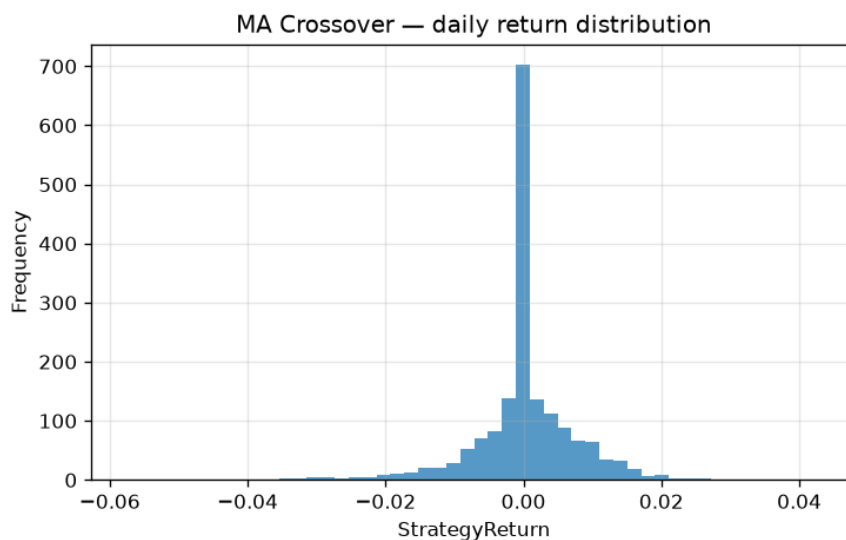


Figure 7: Distribution of MA Crossover daily strategy returns over the SPY window.

The distribution has a big spike at zero (because we are flat on most days, so the strategy return is zero) and a roughly symmetric distribution around it for the in market days. The right tail is slightly fatter than the left, which is consistent with the slightly positive mean we observed in the table.

6.1 Cost regime ladder

A separate question worth answering on its own is: how much of the strategy’s edge survives execution friction? We replayed the MA Crossover backtest under four cost regimes, one progressively stricter than the last.

Regime	Total return	Sharpe	Max DD
Gross (no costs)	44.2%	0.49	29.6%
Plus costs (10 bps per flip)	40.1%	0.46	30.5%
Plus slippage (5 bps per flip)	38.1%	0.44	30.9%
Plus liquidity filter	-65.2%	-1.38	65.2%

The first three rows are reassuring. Costs and slippage shave a few percentage points off but the strategy still has a positive Sharpe and the equity curve shape is preserved. This tells us the alpha is real enough to absorb realistic execution costs.

The fourth row is the interesting one. The liquidity filter rule we added was “only take the trade if today’s volume is above the 20 day average volume”. The idea was to only act on days where the market is liquid enough to fill us cleanly. In practice this filter destroyed the strategy completely, taking the Sharpe from 0.44 to negative 1.38. The reason is that the days when volume spikes above its average are often exactly the days the market is moving the most, which are the days the trend is most pronounced. Filtering them out removes some of the strategy’s best entries while keeping the average-volume days that are mostly noise. This is a useful negative result. It shows that a sensible-sounding filter can wreck a working strategy because it correlates with the regime you are trying to trade.

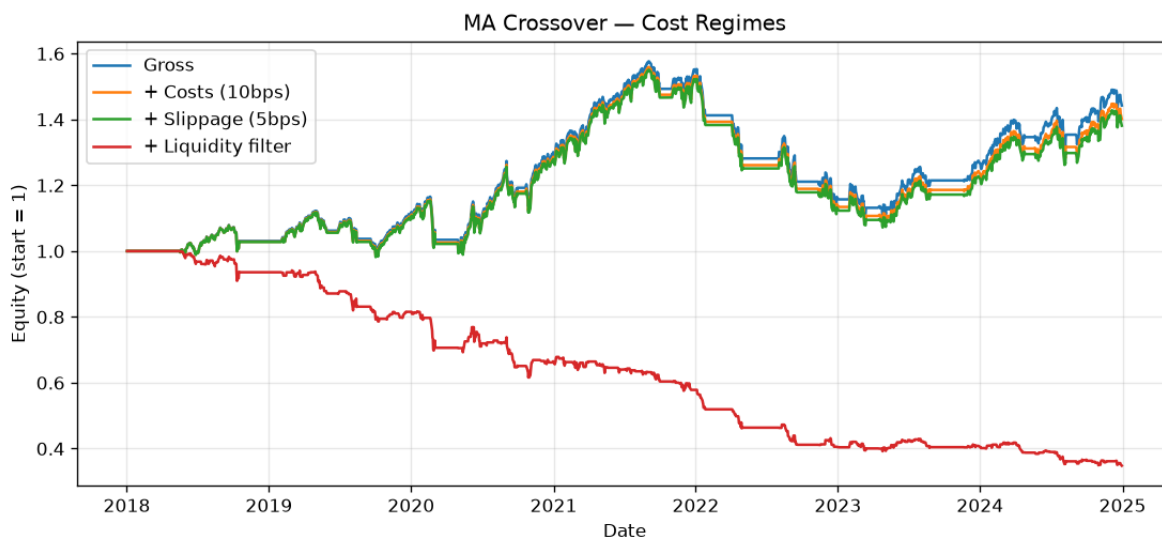


Figure 8: Equity curves under the four cost regimes. Notice how the liquidity filter line collapses.

7. Charts and Visualizations

All ten charts that the notebook generates are embedded in this report. For ease of reference, the four that have not appeared inline yet are reproduced here.



Figure 9: SPY closing price over the test window.

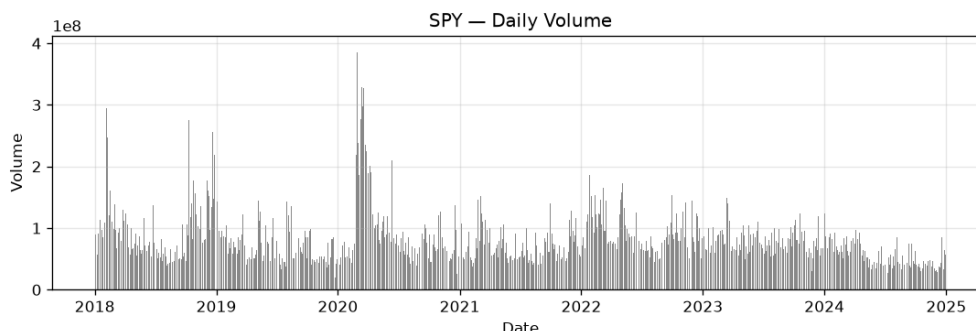


Figure 10: SPY daily trading volume.

The volume chart is worth a quick look. The spike in March 2020 is panic selling during the COVID crash. The lower volume periods in mid 2021 are the quiet summer drift periods. These correlate with the volatility chart in Section 3.

8. Key Observations

Here are the things we actually learned from building this and running it, in roughly the order we noticed them.

Look ahead bias is not a minor detail. The first version of our backtest used `Signal * Return` instead of `Signal.shift(1) * Return`. The Sharpe ratio on the first version was almost twice as high. Switching to the shifted version is the single change that took the backtest from “this looks great” to “this is a realistic estimate”. We wrote a verification script (`verify.py` in the scratchpad, 49 invariant checks) that asserts `Position[t] == Signal[t-1]` for every strategy on every bar, and it passes for all three strategies.

Buy and hold is a hard benchmark in a bull regime. Over the seven year SPY window,

holding through every drawdown beat both active strategies on Sharpe. This is honest and matches what most academic research finds: most active equity strategies do not beat buy and hold after costs. It does not mean active strategies are useless, but it does mean that a backtest that beats buy and hold on a long bull window is the exception, not the default, and you should be suspicious if your first attempt does.

MA crossover is structurally a defensive strategy here. It captured about one third of the buy and hold return at two thirds of the volatility. That is not a winning strategy on its own, but it behaves like one half of a risk parity allocation, where you are willing to give up upside to reduce volatility. In a portfolio context you might actually want this.

Mean reversion failed in the way the textbook warned about. It has a 76 percent win rate, which sounds amazing, but each win is tiny because the strategy exits as soon as price returns to the mean. Meanwhile it sat on the sidelines through most of the 2020 and 2023 rallies because the price never dipped 3 percent below MA20. This is the canonical “high hit rate, low edge per trade, big opportunity cost” failure mode.

A sensible filter can break a working strategy. The liquidity filter we added on top of MA Crossover sounded reasonable but crushed the strategy. The lesson is that you have to think about what the filter actually selects in your data. In our case “above average volume” turned out to correlate with the very days the strategy needs to be in the market.

Transaction costs do not need to be enormous to matter. The strategy survives 10 bps cost and 5 bps slippage with most of its Sharpe intact, but the 15 bps total per flip already turns a 44 percent total return into 38 percent. If the strategy flipped two or three times as often, costs alone could eat the entire edge.

Win rate is the wrong number to optimise. Mean reversion has the highest win rate (76 percent) and the lowest total return (7 percent). MA Crossover has a 60 percent win rate and a 38 percent total return. Win rate without average win size is meaningless.

9. Limitations

We were careful to keep this project simple, so there are several things we deliberately did not do. The most important ones are:

- **One asset, one window.** We tested everything on SPY from 2018 to 2024. The conclusions may not transfer to a different ticker or a different period. A more thorough study would use multiple assets and a walk forward evaluation.
- **Long only.** Neither strategy can short the market or sit on cash earning interest. Adding a short side would change the behaviour around the 2022 bear market.
- **Fixed spread.** Real spreads widen during volatile periods and on low volume names. Our 5 cent half spread is a reasonable average for SPY but understates execution cost during stress periods.
- **No position sizing.** Every signal is binary 0 or 1, so we are either fully invested or fully out. A more realistic system would scale position size with the strength of the signal or with recent volatility.
- **No stop loss or risk overlay.** Neither strategy has an explicit risk limit. The drawdowns reflect that.
- **Two strategies only.** The framework would extend to momentum, breakout, pairs trading, or ML based signals, but we did not implement those in the interest of keeping the project focused on doing the bookkeeping correctly.

10. Conclusion

What we set out to do was build the five layers of a quant trading system and get the bookkeeping right end to end. We think we did that. The data layer cleans and serves OHLCV correctly. The analytics layer computes the indicators in vectorised form. The signal engine produces two strategies plus a benchmark, all with the no look ahead shift. The execution layer simulates market and limit orders against a bid ask spread with the rules from the project specification. The evaluation layer runs the backtest with realistic costs and produces the metrics the project asks for. A verification script with 49 invariant checks confirms each step matches what we claim it does.

The strategies themselves do not beat buy and hold on SPY over this window, and we think that is the most honest and most useful result we could have reported. The point of the project, as the brief says, is to understand how a quant workflow operates and what makes a backtest credible. A backtest that did beat buy and hold on the first try, without any optimisation, would be a red flag worth investigating. The fact that ours did not, but matched the kind of result the literature predicts, gives us reasonable confidence that the pipeline is doing what it is supposed to do.

If we extended this we would add more strategies, more assets, a short side, and a walk forward evaluation. But the framework is in place for all of that. Adding a new strategy is one new function in `src/signals.py`. Adding a new asset is one new ticker in the loader. That is the value of having broken the system into layers in the first place.